

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0558

Candelaria 138 kV
August 22, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0558. The Interconnection Customer proposes a 285.4 MW natural gas combined cycle facility interconnecting at the Candelaria 138 kV substation in the Virginia area. The request was received on December 25, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 231 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,625,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of SPRING GAP TAP – COVANTA SOUTHEASTERN CONNECTICUT COMPANY 138.0 kV, the LITTLEFORK – HOGAN RD 138.0 kV loads to 147.0% of its 271 MVA rating. Project DFAX: 5.2%.

The REACH – CANDELARIA 138.0 kV is overloaded to 139.0% of its 293 MVA emergency rating for the loss of the CLOVER – BEAR 345.0 kV. The project contributes 56.0% DFAX (159.9 MW) to this overload.

ALDERWOOD – WAYNESBORO 138.0 kV: 109.1% loading (243 MVA) following the outage of the REACH SUBSTATION – CANDELARIA 138.0 kV; DFAX = 9.7%.

The BANNING – DESERT OUTPOST 138.0 kV is overloaded to 107.1% of its 190 MVA emergency rating for the loss of the FRANKLIN-ELK GROVE – S TEXA 138.0 kV. The project contributes 11.1% DFAX (31.8 MW) to this overload.

For the loss of RICHLAND – DEKALB 500.0 kV, the DEKALB 230/1 kV XFMR loads to 105.6% of its 1,299 MVA rating. Project DFAX: 6.6%.

For the loss of JONES CHAPEL – DEKALB 500.0 kV, the BLUEBONNET – RUSSELL FORK 230.0 kV loads to 105.3% of its 1,164 MVA rating. Project DFAX: 5.7%.

WAYNESBORO – CENTRAL AUSTIN 138.0 kV L: 104.9% loading (229 MVA) following the outage of the CANDELARIA – REACH 138.0 kV; DFAX = 9.7%.

The COAL MTN – CENTRAL AUSTIN 138.0 kV Line is overloaded to 103.2% of its 224 MVA emergency rating for the loss of the REACH – CANDELARIA 138.0 kV. The project contributes 9.7% DFAX (27.7 MW) to this overload.

COAL MTN SUBSTATION – GRASSMERE 138.0 kV line: 101.0% loading (222 MVA) following the outage of the REACH – CANDELARIA 138.0 kV L; DFAX = 9.7%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LITTLFORK – HOGAN RD 138.0 kV	Rebuild line with higher-capacity conductor	\$23,443,000	20% (\$4,798,000)
REACH – CANDELARIA 138.0 kV	Rebuild line with higher-capacity conductor	\$11,515,000	51% (\$5,908,000)
ALDERWOOD – WAYNESBORO 138.0 kV	Reconductor line	\$1,362,000	17% (\$228,000)
BANNING – DESERT OUTPOST 138.0 kV	Reconductor line	\$3,286,000	13% (\$422,000)
DEKALB 230/1 kV XFMR	Replace transformer with higher-rated unit	\$8,933,000	2% (\$180,000)
BLUEBONNET – RUSSELL FORK 230.0 kV	Reconductor line	\$15,084,000	2% (\$338,000)
WAYNESBORO – CENTRAL AUSTIN 138.0 kV L	Reconductor line	\$1,097,000	17% (\$184,000)
COAL MTN – CENTRAL AUSTIN 138.0 kV Line	Reconductor line	\$1,599,000	17% (\$268,000)
COAL MTN – GRASSMERE 138.0 kV	Reconductor line	\$1,782,000	17% (\$299,000)
	Total allocated to project		\$12,625,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.